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Kartik Loya

Professional Summary

Ph.D. candidate in Mechanical Engineering focused on physics-informed, data-driven modeling and control methods for unconventional robotic platforms, including off-road ground vehicles on deformable terrain and fish-like robots with internal actuation. I genuinely enjoy building and testing robots, and I'm especially interested in autonomy that works outside the lab in messy, real environments. My research blends first principles dynamics, multi-modal sensing, and data-driven system identification to build control-oriented predictive models, and I use optimization and learning based control to translate these models into reliable, real-time robot behavior under uncertainty.

Education

Clemson University — Clemson, USA
Ph.D. in Mechanical Engineering — 2021 - 2026 *(expected)*
- GPA: 4.00/4.00

M.S. in Mechanical Engineering — 2018 - 2021
- GPA: 3.87/4.00.

Maharashtra Institute of Technology, Pune University — Pune, India
B.E. in Mechanical Engineering — 2014 - 2018

Peer Reviewed Publications

1. Under Preparation
   - **Loya K**, Chivkula, P, Tallapragada P. **Hierarchical Koopman Operator-Based Control of a Fish-Like Robot with Internal Reaction Wheel Dynamics.**
   - **Loya K**, Chivkula, P, Varikuti R, Tallapragada P. **Hybrid RL–PID Control of a Biomimetic Underwater Robot Under Flow Disturbances**
2. Published
   - **Loya K**, Tallapragada P. **Koopman Operator Framework for Modeling and Control of Off-Road Vehicle on Deformable Terrain.** ***submitted to ASME Journal of Autonomous Vehicles and Systems. 2025***
   - Chivkula, P, **Loya K**, Tallapragada P. **Agile 3-D Motion Primitives in an Underactuated Fish-like Robot.** ***submitted to IEEE Robotics and Automation - Letters. 2025***
   - **Loya K**, Tallapragada P. **Parametric roll oscillations of a hydrodynamic Chaplygin sleigh.** *Nonlinear Dynamics 111 (22), 20699-20713.*, 2023.
   - **Loya K**, Tallapragada P. **Loya, Kartik, and Phanindra Tallapragada. "Online learning of Koopman operator using streaming data from different dynamical regimes." IFAC-PapersOnLine 58.28 (2024): 90-95.**
   - **Loya K**, Buzhardt J, Tallapragada P. **Koopman operator-based predictive control uses a data archive of observables.** *ASME Letters in Dynamic Systems and Control 3 (3)*, 2023.
   - **Loya K**, Tallapragada P. **Stabilization of an Inverted Pendulum on a Nonholonomic System.** *IFAC-PapersOnLine 55 (37), 764-769*, 2022.

Research Area

Data-driven modeling and control strategies for understanding locomotion in complex robotic systems, optimal control and trajectory optimization, Unconventional robot dynamics (bio-inspired underwater robots, off-road vehicle autonomy), Nonholonomic/underactuated systems, Physics-informed learning Koopman/Operator-theoretic modeling.

Research Experience

Virtual Prototyping of Ground Vehicle Systems Center (VIPR-GS)
Graduate Research Assistant September 2024 - Present

Advisor: Prof. Phanindra Tallapragada
Data-driven Koopman Control-Oriented Learning Framework for Improved Autonomous Offroad Mobility

- Developed a physics-informed, data-driven Koopman learning framework for off-road vehicle mobility on deformable terrain, enabling control-oriented prediction without expensive online terramechanics evaluations.
- Identified soil-specific lifted linear predictors using scalable (recursive) subspace identification, with data prioritization to capture regime changes from large simulation datasets.
- Integrated the learned predictor into constrained MPC to achieve reliable real-time tracking of aggressive trajectories under steering and torque limits.

Applied Dynamical Systems Lab
Graduate Research Assistant August 2021 - May 2024

Advisor: Prof. Phanindra Tallapragada

1. Modelling and control of data-driven dynamical systems
   - Developed a linear predictor for high-dimensional nonlinear systems using Koopman operator and subspace identification methods.
   - Demonstrated effective control in systems like the hydrodynamic Chaplygin sleigh by applying adaptive control for accurate system stabilization and trajectory tracking.
   - Created an online learning framework for updating Koopman operators from streaming data across different dynamical regimes. This method efficiently handles large datasets by adaptive data pruning and quantifies uncertainty in Koopman-based modeling frameworks by using Gaussian processes as basis functions.
2. Investigated the roll dynamics of a hydrodynamic Chaplygin sleigh, analyzing how parametric oscillations and nonholonomic constraints affect the stability of bioinspired underwater robots by simplifying the complex nonlinear model into a linear Mathieu oscillator.
3. Designed and developed a fish-inspired robot with internal rotor actuation, and currently using data-driven methods (Koopman operator or RL) for its modeling and control.

Virtual Prototyping of Ground Vehicle Systems Center (VIPR-GS)
Graduate Research Assistant January 2020 - August 2021

Advisor: Dr. Mohammad Naghnaeian

- Energy management of multi-scale vehicle fleets
- Designed energy-efficient planning and control strategies for UAV fleets in resource-constrained environments.
- Used MPC to enhance UAV energy efficiency, factoring in disturbances, wind effects, and morphed UAV for improved performance.

COMPETITIONS AND ROBOTICS CHALLENGES

AI for Industry Challenge - Intrinsic × Open Robotics | Team ASIMOV May 2026 - August 2026

- Ranked 21st globally and advanced to Phase 1 (Top 30 of 160+ teams) in a worldwide robotics competition focused on autonomous manipulation of deformable cables and precision connector insertion.
- Developed the manipulation solution using Intrinsic Flowstate, integrating Vision-Language-Action (VLA) models, reinforcement learning, and OpenCV-based perception for task planning, visual understanding, and robot control.
- Designed and tested reusable robot skills for perception and manipulation within the Flowstate workflow, combining learning-based approaches with task-specific robotic behaviors.

TECHNICAL SKILLS

Programming Languages: MATLAB, Python, C++, Julia-(beginner).

Software: Mathematica, Solidworks, ROS, ROS2, Pytorch, Tensorflow, GPOPS-II, MuJoco.

WORK EXPERIENCE

Thermax Ltd. | Pune, India October 2017 - May 2018

- Surveyed and optimized the steel utilization in the plant.
- Developed a Python program for remote online performance analysis for the Vapour Absorption chiller plants.
- **Capstone Project:** Design and manufacturing of boiler panel cutting machine to replace the existing flame cutting method used in repairing boilers to avoid the development of heat-affected zones.

Elite Techno Group | Pune, India June 2016

- Designed ATV on Solidworks and performed crash analysis on it.
- Manufactured the ATV and tested its performance on a road-track.

INVITED / CONFERENCE TALKS & POSTERS PRESENTATION

- Modeling, Estimation and control Conference (MECC)
  1. 2022 "Stabilization of an Inverted Pendulum on a Nonholonomic System"
  2. 2023 "Koopman Operator Based Predictive Control With a Data Archive of Observables "
  3. 2024 "Online learning of Koopman operator using streaming data from different dynamical regimes "
- SIAM Conference on Applications of Dynamical Systems (DS25) -Talk: Model-based learning of Koopman operator for complex mechanical systems (May 2025).
- ISTVS 2025 (55th Conference) - Talk: Data-driven Koopman control-oriented learning framework for improved autonomous off-road mobility (Oct 9, 2025).
- ICRA 2025 - Poster: "Batch learning/updates of Koopman operators from streaming off-road vehicle data with terrain dynamics (Summer 2025).
- ICRA 2025 - Late Breaking Results: "Learning PID Gains for Planar Tracking of Bio-Inspired Swimming Robots Using Simplified Models"
- Clemson VIPR Annual Meeting 2025 - Poster Presentation: "Koopman-Based Modelling and Control of Off-Road Vehicles on Deformable Terrain"
- Clemson Graduate Student Seminar (GSR) — Research talks
  1. 2023 "Parametric Roll Oscillation of a Hydrodynamic Chaplygin Sleigh"
  2. 2025 "Data-driven Koopman Control-Oriented Learning Framework for Improved Autonomous Off-road Mobility."

Graduate Coursework

Intermediate Dynamics	Nonlinear Dynamics & Chaos
Robot Multibody Dynamics	Advanced Nonlinear Dynamics
Advance Engineering Mathematics	Linear Analysis
Control Systems Engineering	Advanced Nonlinear control
Random Processes and Application	Applied Optimal Control
Networked Dynamical systems	Advanced Estimation
Control of Cyber-Physical Systems	Autonomy: Science & Systems
Reinforcement Learning	Advanced Machine Learning
Theory of Elasticity	